SearchAgent: Behavioral Calibration of LLM Semantic Priors Supplementary Material

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Abstract

Extended technical definitions, exact evaluation protocols, per-dataset experiments, robustness analyses, and complete LLM prompts accompanying the main paper. This PDF is a standalone companion to the anonymized paper and code, not part of the 12-page submission PDF.

Navigation. Sections A–G retain the complete supporting material; the compact appendix in the submission PDF maps the eight-page paper's original cross-references to these sections. References to equations, figures, and tables in the paper are imported from the companion paper source for correct numbering (compile the submission first).

A Behavioral Fidelity Metrics

This appendix provides the definitions of the standard metrics used in Section 4.9. Let $\mathcal{Y}$ and $\mathcal{Y}^\star$ denote the generated and real session sets, respectively.

Distributional discrepancies. We compare the clicked-rank and session-length distributions using Jensen–Shannon divergence:

$$\begin{aligned} \mathrm{JSD}(P\|Q) &= \frac{1}{2}\,\mathrm{KL}(P\|M) + \frac{1}{2}\,\mathrm{KL}(Q\|M), \\ M &= \frac{P+Q}{2}. \end{aligned} \tag{1}$$

Accordingly, the two distributional discrepancies are

$$\begin{aligned} D_{\text{rank}} &= \mathrm{JSD}\left(P^{\text{gen}}_{\text{rank}} \middle\| P^{\text{real}}_{\text{rank}}\right), \\ D_{\text{length}} &= \mathrm{JSD}\left(P^{\text{gen}}_{\text{length}} \middle\| P^{\text{real}}_{\text{length}}\right). \end{aligned} \tag{2}$$

Behavioral rate gaps. For a session set $\mathcal{S}$, let $\mathrm{CPS}(\mathcal{S})$ denote the mean number of clicks per session, $\mathrm{AR}(\mathcal{S})$ the abandonment rate, and $\mathrm{RR}(\mathcal{S})$ the reformulation rate. Their discrepancies are defined as

$$\begin{aligned} D_{\text{click}} &= \left|\mathrm{CPS}(\mathcal{Y}) - \mathrm{CPS}(\mathcal{Y}^\star)\right|, \\ D_{\text{abandon}} &= \left|\mathrm{AR}(\mathcal{Y}) - \mathrm{AR}(\mathcal{Y}^\star)\right|, \\ D_{\text{reform}} &= \left|\mathrm{RR}(\mathcal{Y}) - \mathrm{RR}(\mathcal{Y}^\star)\right|. \end{aligned} \tag{3}$$

Replay metrics. For each replayed session y, let $C^{\text{gen}}_y(r)$ and $C^{\text{real}}_y(r)$ denote the numbers of generated and real clicks at rank r. The matched click count is

$$H_y = \sum_r \min\left\{C^{\text{gen}}_y(r), C^{\text{real}}_y(r)\right\}. \tag{4}$$

We compute micro-averaged replay precision, recall, and F1 as

$$\begin{aligned} \mathrm{Prec} &= \frac{\sum_y H_y}{\sum_y \sum_r C^{\text{gen}}_y(r)}, \\ \mathrm{Rec} &= \frac{\sum_y H_y}{\sum_y \sum_r C^{\text{real}}_y(r)}, \\ \mathrm{F1} &= \frac{2\,\mathrm{Prec}\,\mathrm{Rec}}{\mathrm{Prec}+\mathrm{Rec}}. \end{aligned} \tag{5}$$

When a denominator is zero, the corresponding score is defined as zero.

Finally, abandonment agreement is

$$\mathrm{AA} = \frac{1}{|\mathcal{Y}^\star|}\sum_y \mathbf{1}\left[A^{\text{gen}}_y = A^{\text{real}}_y\right], \tag{6}$$

where $A_y = 1$ indicates that session y contains no click.

B SERP Construction, Field Provenance, and Missingness

Public click logs do not necessarily provide complete or human-readable SERPs. We therefore represent each result using the canonical record (ρ, title, snippet, url, m^{text}) and preserve the provenance of every field throughout the pipeline. Missing titles, snippets, URLs, and queries are never synthesized by an LLM. Table 1 summarizes the construction procedure and environment behavior for each dataset.

B.1 Deterministic Field Completion

For TREC and ORCAS, document lookup is exact and is completed before any split-specific feature estimation. A nonempty corpus title is copied verbatim after Unicode normalization. If no title is available, the field remains empty and its availability mask is set accordingly; the document identifier is never substituted for the missing title. Snippets are constructed by segmenting the document body into sentences, scoring each sentence according to its BM25 overlap with the current query, and selecting the highest-scoring contiguous window subject to a limit of 320 Unicode characters. Ties are resolved in favor of the earliest document position. If the document body is unavailable, the snippet remains empty. A corpus URL is retained after credentials and fragments have been removed. When only an opaque identifier is available, it is serialized under `opaque_document_id` and excluded from semantic scoring. Result-level text availability is defined as

$$m^{\text{text}}_{t,j} = \mathbf{1}\left[\text{title} \vee \text{snippet} \vee \text{URL is non-opaque}\right]. \tag{7}$$

The candidate-attractiveness interface receives a separate availability mask for each field. If every candidate on a SERP satisfies $m^{\text{text}} = 0$, the semantic candidate prior is set exactly to $1/m_t$, thereby preventing the LLM from assigning semantic meaning to opaque identifiers. The macro-action prior may still condition on observable structural information, including the number of ranked results, the current page, interaction history, and cognitive state, but it is explicitly informed that the SERP contains no usable text through the `text_blind=true` flag.

B.2 Retrieval for Query-Changing Actions

The treatment of query-changing actions follows the retrieval resources available for each dataset. For the TREC collections, a generated query is executed against the frozen corpus associated with

Table 1: SERP-field provenance and environment behavior. Corpus join denotes an exact document-identifier lookup in the corresponding frozen collection and does not involve a contemporary search engine.

Dataset	Logged query and ranking	Title, snippet, and URL	SERP after a generated query
TREC12	Session XML and logged ranked document identifiers; entries with malformed ranks are discarded.	Logged display text is retained when available. Missing fields are recovered through exact joins to ClueWeb09. Titles and URLs are copied from corpus records, and snippets are extracted deterministically from document text.	BM25 retrieval over the frozen ClueWeb09 index, followed by the same field-extraction procedure and a fixed page size of ten results.
TREC13/14	Session records and logged ranked document identifiers.	Logged display text is retained when available. Missing fields are recovered through exact joins to ClueWeb12 and processed using the same procedure as for TREC12.	BM25 retrieval over the frozen ClueWeb12 index. The index version, analyzer configuration, and tie-breaking seed are fixed before evaluation.
ORCAS	Official query–document click pairs evaluated under a single-query protocol.	Each document identifier is joined to the MS MARCO document corpus, which provides the URL, title, and body. Snippets are extracted deterministically from the body with respect to the query.	Evaluation is conducted on the fixed query-conditioned candidate pool.
Yandex	Anonymized user, session, query-term, URL, domain, rank, and click identifiers.	URL and domain identifiers are retained as opaque categorical values. They are not decoded or converted into titles, snippets, domain names, or URLs, and $m^{\text{text}} = 0$.	Query-changing actions follow the recorded session trajectory.

the corresponding benchmark. TREC12 uses ClueWeb09, whereas TREC13 and TREC14 use ClueWeb12. Retrieval is performed using a reproducible BM25 index with lowercase Unicode normalization, collection-standard tokenization, $k_1 = 0.9$, $b = 0.4$, and stable document-identifier tie breaking. The index contains neither relevance judgments nor test-set click information, and the same retrieval configuration is shared across all methods and random seeds. Live search APIs are excluded because their outputs may change over time and may introduce information beyond the benchmark collections.

ORCAS is evaluated under a single-query protocol using its fixed query-conditioned candidate pool. Query-changing actions therefore do not trigger retrieval of a new SERP in this setting. This restriction preserves the observed candidate context and avoids introducing retrieval results that are not supported by the evaluation protocol.

The Yandex logs do not provide recoverable document content or a searchable document collection against which newly generated queries can be executed. Consequently, query-changing actions do not produce independently retrieved SERPs. Instead, evaluation follows the query transitions and ranked-result states recorded in the original sessions. This structured replay preserves the observable progression of each logged session while avoiding unsupported reconstruction of document semantics or retrieval outcomes.

C Reproducibility Configuration

This section specifies the data partitions, model configuration, optimization procedure, baseline adapters, compute accounting, and uncertainty analysis used in the experiments. Unless otherwise stated, stochastic methods are evaluated over three independent runs with seeds $\{42, 123, 456\}$. Each seed controls parameter initialization, minibatch ordering, dropout, action sampling, dwell-time noise, and method-specific sources of randomness. Finite-label LLM requests use fixed prompts, decoding settings, and validation rules, while the request and response metadata returned by the provider are retained for each uncached call. A single dataset partition, generated with split seed 2027, is shared across the three runs. Reported values are means and standard deviations over these runs unless noted otherwise.

C.1 Data Partitions and Leakage Prevention

We first remove malformed records, retain at most ten visible results per SERP, and truncate session logs to a maximum of twenty environment steps. Query canonicalization applies Unicode NFKC normalization, lowercasing, whitespace collapse, and removal of terminal punctuation. All records connected through the dataset-specific grouping keys are assigned to the same partition before feature fitting, prompt caching, or LLM scoring. The resulting train, validation, and test partitions are summarized in Table 2.

Persistent user identifiers are not available for the TREC collections. Consequently, user-level separation cannot be verified and is not inferred from session identifiers. For Yandex, we construct connected components in the bipartite graph linking user identifiers and canonical queries. Assigning each component as an indivisible unit prevents a query shared by multiple users from creating an indirect connection across partitions. Components exceeding 2% of the sampled corpus are retained intact and assigned using a deterministic, size-balanced procedure. Feature-normalization statistics are estimated from the training partition only. The validation partition is used for model selection and hyperparameter selection, and the held-out test partition is evaluated once for each random seed.

C.2 Model, Architecture, and Optimization

The primary experiments request deepseek-v4-pro through the provider's OpenAI-compatible endpoint. The cross-backbone analysis additionally requests gpt-5.6-sol through the OpenAI Responses API and claude-sonnet-5 through the Claude Messages API. The requested model identifiers, returned identifiers, and provider interfaces used in each run are recorded in the experiment manifest.[1] Because these models are provider hosted, the experiments do not assume access to model weights, tokenizer files, numerical precision, quantization settings, CUDA versions, or accelerator configurations.

Tools, web browsing, external retrieval, and cross-request conversational state are disabled. Requests use structured JSON outputs and are limited to at most 8,192 serialized input tokens according to the provider-side request accounting. Finite-label probability requests allow at most 256 output tokens, while query generation is limited to 32 whitespace-delimited tokens. For every uncached request, we retain the requested and returned model identifiers, request ID, UTC timestamp, canonical prompt hash, response hash, provider-reported token usage, retry count, and parser status. Appendix G describes the shared output schemas, validation rules, and failure-handling procedure.

[1] https://api-docs.deepseek.com/quick_start/pricing; https://developers.openai.com/api/docs/models/gpt-5.6-sol; https://www.anthropic.com/news/claude-sonnet-5.

Table 2: Train, validation, and test partitions together with the grouping units used to reduce cross-partition leakage. Counts denote sessions for TREC and Yandex and unique queries for ORCAS.

Dataset	Train	Validation	Test	User ID	Grouping unit and overlap check
TREC12	60	19	19	unavailable	Connected grouping by session ID and canonical query; no session or exact canonical-query overlap across partitions.
TREC13	80	26	27	unavailable	Connected grouping by session ID and canonical query; no session or exact canonical-query overlap across partitions.
TREC14	754	251	252	unavailable	Connected grouping by session ID and canonical query; no session or exact canonical-query overlap across partitions.
Yandex	40,000	5,000	5,000	available	Connected components induced by user ID and canonical query; no user, session, or exact canonical-query overlap across partitions.
ORCAS	80,000	10,000	10,000	unavailable	Canonical query; no exact query overlap across partitions. No session-level or user-level separation is claimed.

Table 3: Fixed state-transition and session-control parameters used in the reported experiments.

Component	Setting	Value
Session budget	Maximum environment steps; maximum SERP pages; results per page	20; 3; 10
ORCAS budget	Single-query protocol with a click or no-click decision	1 step
Memory	Most recent complete interaction records retained in the prompt	8 records
Numerical stability	Probability smoothing in Eqs. (17)–(18)	$\epsilon = 10^{-8}$
Dwell model	Base dwell times $B(1{:}5)$ in seconds, piecewise-linear interpolation for noninteger s, and log-normal noise σ_d	$[5, 15, 35, 70, 120]$; 0.35
Click-state update	Increment $\Delta(\text{clar}, \text{know}, \text{frus}, \text{focus})$ for normalized utility $u \in [0, 1]$	$(0.08u, 0.12u, 0.05(1 - u) - 0.08u, 0.03)$
Skip / next page	Fixed cognitive-state increments	$(-0.01, 0, 0.03, -0.02)$ / $(-0.02, 0, 0.05, -0.04)$
Reformulate / new query	Fixed cognitive-state increments	$(0.06, 0, -0.02, 0.04)$ / $(-0.03, 0, -0.01, -0.06)$

Table 4: Deterministic mappings from baseline outputs to the unified environment action space.

Method	Native decision form	Adapter used in the evaluation harness
ReAct	Thought, action, and tool-command sequence	The final `CLICK(i)`, `SKIP`, `NEXT`, query-change, or stop command is executed. Intermediate thoughts are ignored by the environment.
PolySkill	Selected skill and associated arguments	Click and navigation skills map to the corresponding environment actions. Refinement and new-search skills are distinguished using the declared transition mode, and the finish status determines satisfied or unsatisfied stopping.
Reflexion	Reflection followed by a final command	Only the command produced after reflection is executed. It is parsed using the same final-command rules as ReAct.
Chain-of-Thought	Rationale followed by a final action label	The final tagged label A–G is mapped through ϕ. Untagged rationale text has no direct effect on the environment action.
Tree of Thoughts	Candidate branches and a selected leaf	Up to six candidate branches may be scored internally. The final tagged command associated with the selected leaf is executed.
UserSimCRS v2	Inspect, accept, reject, request, and end acts	Accept or inspect with an item maps to click; reject or continue maps to skip; request-more maps to next page; clarify maps to reformulation; and end is mapped to a stopping action using the emitted satisfaction indicator.
CSHI	Click, continue, refine, switch, and finish acts	Click, continue, refine, and switch map to click, skip, reformulation, and new-query actions, respectively. Finish is mapped to satisfied or unsatisfied stopping using the method's satisfaction score with threshold 0.5.
USimAgent	Query, click, browse, or stop event	Native click, browse, query, and stop events map to their corresponding unified actions. A query change with normalized Levenshtein distance below 0.5 is categorized as reformulation; otherwise, it is categorized as a new query.
BASES	Whole-session event sequence	Predicted events are supplied to the environment sequentially. Missing intermediate clicks do not create additional actions. Query changes use the same normalized edit-distance rule as USimAgent, and terminal utility of at least 3 maps to satisfied stopping.
Agent4Rec	Item interaction and exit intent	Item selection maps to click; browsing maps to skip or next page according to whether a page boundary is reached; preference refinement maps to reformulation; goal shift maps to a new query; and the exit rating determines the stopping type.
SimUSER	Structured user act	Item choice, continue, request-more, refine, switch-goal, and finish are mapped to click, skip, next page, reformulation, new query, and stopping, respectively.
ConvSearch-R1	Search, reformulate, browse, click, and finish acts	Native browse, click, and finish acts map directly. A generated query retaining at least half of the canonical tokens from the current query is categorized as reformulation; otherwise, it is categorized as a new query. The finish output includes the stopping satisfaction state.

The state vector in Eq. (15) contains 17 dimensions. The behavioral calibrator f_θ is a 17–64–32–7 multilayer perceptron with LayerNorm, GELU activations, and dropout 0.1 after each hidden layer. The reliability gate is the seven-parameter constrained generalized linear model defined in Eq. (20), with $\alpha_{\max} = 2.0$. The examination network has dimensions 5–16–1 and uses GELU followed by softplus. The attractiveness network has dimensions 3–32–16–1 and uses LayerNorm and GELU. Across these learned components, the implementation contains approximately 4.5K trainable parameters.

Training minimizes the equally weighted mean of the action and click negative log likelihoods. We use AdamW with learning rate 3×10^{-4}, $\beta = (0.9, 0.999)$, and weight decay 10^{-4}. Gradients are clipped to a maximum norm of 1.0. The action and click batch sizes are 512 and 256, respectively. Training proceeds for at most 100 epochs and terminates after ten consecutive validation epochs without an improvement of at least 10^{-4} in the equally weighted validation action and click NLL. Evaluation uses the checkpoint with the best validation objective rather than the final training epoch. The LLM parameters remain frozen, and LLM scores for logged training states are computed before calibrator training.

Hyperparameters are selected once from 36 randomly sampled configurations using a pooled validation objective in which each dataset receives equal weight. The search space comprises learning rates $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$, hidden-layer widths $\{(32, 16), (64, 32), (128, 64)\}$, dropout rates $\{0, 0.1, 0.2\}$, weight-decay values $\{0, 10^{-5}, 10^{-4}\}$, $\alpha_{\max} \in \{1, 2, 4\}$, intent momentum $\lambda_\pi \in \{0.6, 0.8, 0.9\}$, and memory lengths $\{4, 8, 12\}$. The selected configuration uses hidden widths $(64, 32)$, dropout 0.1, $\alpha_{\max} = 2$, $\lambda_\pi = 0.8$, and eight memory entries. The same selected configuration is used across datasets and is fixed before test evaluation.

The intent-transition targets follow the coordinate order exploratory, focused, and transactional. The targets are $(0.20, 0.60, 0.20)$ for click, $(0.45, 0.45, 0.10)$ for skip, $(0.55, 0.35, 0.10)$ for next page, $(0.25, 0.65, 0.10)$ for reformulation, $(0.65, 0.25, 0.10)$ for a new query, $(0.10, 0.75, 0.15)$ for satisfied stopping, and $(0.40, 0.45, 0.15)$ for unsatisfied stopping. Cognitive-state values are clipped to $[0, 1]$ after each update.

C.3 Unified Baseline Action Adapters

The evaluation harness requires each baseline to return one final, machine-readable command at every environment step. Intermediate reasoning, reflection text, and tree-search branches are treated as internal computation and are not counted as environment actions. When a returned command is invalid, the method is given one retry with the same interaction context. If the second response remains invalid, the environment applies `stop_unsatisfied`. Table 4 specifies the deterministic mapping from each method's native output format to the unified action space.

C.4 Equal-Supervision Control Protocol

The primary baseline comparison retains the training regime originally associated with each method. Although this reflects their standard use, differences in supervision and model selection can confound comparisons with SearchAgent. We therefore conduct an additional equal-supervision control using the common train and validation partitions described in Appendix C. All comparators receive the same observable logged transitions relevant to their prediction targets, the same feature availability masks, and the same 36-trial model-selection budget. Cached LLM outputs are shared when the comparator uses an LLM interface. No method receives test-partition labels, test-set marginals, or aggregate behavioral statistics.

For neural comparators whose architectures permit direct capacity matching, the number of trainable parameters is kept close to that of the approximately 4.5K-parameter learned component in SearchAgent. Classical click models retain their lower-capacity parameterizations rather than being enlarged solely to match a parameter count; all implementation totals are reported in Table 5. Validation objectives remain aligned with the outputs predicted by each model, and the held-out test partition is not used for architecture or hyperparameter selection.

The raw LLM prior + plain MLP comparator concatenates the centered LLM log-probabilities with the 17-dimensional observable state and directly learns action and clicked-item logits. It does not include the reliability gate or the prior-orthogonality constraint. The behavior-cloning MLP predicts the same hierarchical action and clicked-item labels from the observable numerical state without an LLM prior or a specialized behavioral decomposition. The PBM and DBN comparators model rank-conditioned clicks and are paired with a first-order controller for query reformulation and session termination. The neural click–session model jointly predicts click behavior and session continuation from visible result features and interaction history. The calibrated USimAgent and BASES variants retain their frozen native scores and fit a supervised residual adapter using the same training partitions and validation budget.

C.5 LLM API Calls, Caching, and Compute Accounting

At each SearchAgent step, the macro-action and visible-SERP quality interfaces are evaluated. Candidate attractiveness is evaluated only when the macro-action requires selecting a result, and post-click utility is evaluated only after a result has been selected. Query generation is invoked only for a query-changing action in an environment that supports executable text queries. No evaluated method is permitted to exceed six logical LLM interface calls per environment step, although methods may use fewer calls.

Each request is serialized as canonical UTF-8 JSON and indexed in the cache by a SHA-256 hash over the complete prompt bytes, interface name, provider, requested model identifier, structured-output schema version, and decoding controls. A cache entry stores either the validated probability vector or the validated generated-query bytes together with the associated provider metadata. Because the complete serialized state contributes to the key, a change in any prompt field yields a different cache entry. Evaluation labels and population-level test statistics are not included in the request, cache key, or cached output. Cache hits are counted as logical interface calls but do not produce additional billed API requests.

The released accounting procedure derives monetary cost from the recorded request-level usage and the provider price schedule retained for the run period, rather than embedding a fixed historical total in the evaluation code. Runtime for the locally trained reliability, residual, examination, and attractiveness components is

Table 5: Equal-supervision control design and implementation budgets. Parameter counts denote the trainable parameters used in the reported implementations. All methods use the same data partitions and a 36-trial validation budget; supervision and validation objectives are matched to the outputs predicted by each model.

Method	Supervision and control	Params	Trials	LLM cap	Validation target
SearchAgent	Logged macro-actions and clicked items; cached outputs from the five LLM interfaces; structured observable-state masks	4.5K	36	6	Joint action/click NLL
Raw LLM prior + plain MLP	Same action and clicked-item labels, observable state, and cached LLM scores; unconstrained direct logits	4.7K	36	6	Joint action/click NLL
Behavior cloning MLP	Same action and clicked-item labels and observable numerical state; no LLM prior or specialized decomposition	5.1K	36	0	Joint action/click NLL
PBM + Markov session	Logged clicks and ranks for the click model; logged reformulation and stopping labels for the session controller	1.2K	36	0	Click and session NLL
DBN + session controller	Logged click sequences and session labels with a dynamic satisfaction state	2.8K	36	0	Click and session NLL
Neural click–session model	Visible result features, interaction history, action labels, and clicked-item labels; generic joint network	5.4K	36	0	Joint action/click NLL
USimAgent + calibration	Frozen native scores with a supervised residual adapter fitted on the common training partitions	4.6K	36	6	Joint action/click NLL
BASES + calibration	Frozen native scores with a supervised residual adapter fitted on the common training partitions	4.6K	36	6	Joint action/click NLL

Table 6: API accounting for the runs used in the reported evaluation. Logical calls equal billed requests plus cache hits. Token totals, expressed in millions, are obtained from provider-reported usage metadata. Monetary cost is computed using the provider price schedule recorded for the corresponding run period. Local computation is accounted for separately.

Workload	Logical calls	Billed requests	Cache hits	Input tok. (M)	Output tok. (M)	Cost (USD)
Logged-state prior precomputation	1,824,600	632,410	1,192,190	428.7	31.6	$104.82
SearchAgent generation and replay	3,962,840	1,284,900	2,677,940	1,086.4	72.8	$298.47
Cross-backbone robustness	2,418,300	1,604,760	813,540	1,472.6	84.3	$1,746.35
Baseline generation and replay	6,744,120	2,298,670	4,445,450	1,932.8	156.7	$884.91
Total	**14,949,860**	**5,820,740**	**9,129,120**	**4,920.5**	**345.4**	**$3,034.55**

Table 7: Paired PFD differences relative to the strongest baseline on each dataset. Positive Δ indicates lower PFD for SearchAgent. Intervals use paired bootstrap, and p values use paired randomization tests with Holm adjustment.

Dataset	Comparator	Δ PFD	Paired-bootstrap 95% CI	Holm-adjusted p
TREC12	BASES	0.55	[0.08, 1.06]	0.044
TREC13	ConvSearch-R1	0.19	[−0.03, 0.43]	0.118
TREC14	Tree of Thoughts	1.88	[0.51, 3.31]	0.020
Yandex	ConvSearch-R1	2.27	[0.35, 4.37]	0.036
ORCAS	BASES	0.49	[0.13, 0.88]	0.018
Macro relative reduction	Dataset-specific best	**5.2%**	**[2.7%, 7.6%]**	**0.002**

recorded separately from provider inference and is not included in the API-cost totals.

C.6 Paired Uncertainty Analysis

For each dataset and comparator pair, uncertainty is estimated from the held-out evaluation units and the three matched random seeds. We draw 10,000 bootstrap samples with replacement, using complete sessions as the resampling unit for TREC and Yandex and complete queries for ORCAS. The same sampled unit indices are applied to SearchAgent and the corresponding comparator. Within each bootstrap replicate, PFD is recomputed from its available components $\mathcal{M}_d$ (five for session datasets, two for ORCAS) for each matched seed pair, and the resulting paired differences are averaged across seeds $\{42, 123, 456\}$.

We report percentile intervals for

$$\Delta\text{PFD} = \text{PFD}_{\text{baseline}} - \text{PFD}_{\text{ours}}, \tag{8}$$

so that positive values indicate a lower PFD for SearchAgent. These intervals quantify variation over held-out evaluation units conditional on the reported training runs; three seeds do not by themselves support a separate asymptotic estimate of training-seed uncertainty.

As a complementary distribution-free check, we perform a paired randomization test. For each held-out unit, the generated trajectories of the two methods are exchanged with probability 0.5, after which PFD is recomputed. This procedure is repeated 10,000 times. Dataset-level two-sided p values are adjusted across the five comparisons using the Holm–Bonferroni procedure. The resampling seed is 314159 and is distinct from the dataset-partition and model-training seeds.

The pooled comparison favors SearchAgent. Four of five bootstrap intervals exclude zero, and the corresponding paired randomization tests retain significance after Holm adjustment; TREC13 remains inconclusive. The bootstrap intervals measure variation across held-out evaluation units, conditional on the reported training runs, whereas the randomization tests assess method differences under paired exchangeability. This distinction supports an average population-level gain without implying equally strong evidence on every dataset.

Table 8: Normalized clicked-rank discrepancy across five datasets. Lower is better. Values are mean ± standard deviation over seeds 42, 123, and 456; best and second-best results are bold and underlined, respectively.

Method	TREC12	TREC13	TREC14	Yandex	ORCAS	Avg. Rank
SearchAgent (ours)	**7.08±3.16**	**3.93±1.18**	**21.59±6.91**	**61.67±19.06**	**3.16±1.75**	**1.00**
ReAct	20.76±2.91	13.16±2.11	52.82±6.52	117.57±15.35	10.42±3.68	12.00
PolySkill	15.90±4.48	9.57±1.81	41.97±7.36	103.07±22.95	6.85±3.31	10.00
Reflexion	14.47±3.36	8.18±1.33	39.72±5.09	101.47±12.92	5.72±2.78	6.00
Chain-of-Thought	20.82±6.25	14.41±4.02	57.05±14.24	122.04±18.74	11.79±4.25	13.00
Tree of Thoughts	15.96±2.20	9.61±1.99	42.13±5.65	103.46±13.07	6.88±1.99	11.00
UserSimCRS v2	15.85±4.87	9.53±2.32	41.80±6.92	102.67±15.31	5.81±2.14	9.00
CSHI	15.79±4.47	9.49±1.54	41.64±6.91	102.27±14.72	5.78±3.05	8.00
USimAgent	11.87±1.89	8.05±1.28	36.89±8.59	79.33±10.12	4.51±1.78	3.00
BASES	8.75±2.20	4.94±2.06	26.92±4.91	76.77±17.25	3.85±1.57	2.00
Agent4Rec	12.85±2.05	8.14±2.55	37.22±5.92	88.77±14.29	5.68±2.83	5.00
SimUSER	14.52±2.19	8.37±2.91	39.88±6.77	101.87±13.29	5.75±2.99	7.00
ConvSearch-R1	12.79±3.34	8.10±2.19	37.05±7.60	79.73±13.52	4.68±1.74	4.00

Table 9: Normalized click-volume discrepancy across five datasets. Raw click volume is measured in clicks per session (clicks per query for ORCAS); entries are dimensionless bootstrap-normalized gaps. Values are mean ± standard deviation over three runs (seeds 42, 123, and 456). Lower is better.

Method	TREC12	TREC13	TREC14	Yandex	ORCAS	Avg. Rank
SearchAgent (ours)	**9.88±1.63**	**7.77±1.55**	**28.58±6.19**	**65.80±13.01**	**7.66±2.57**	**1.00**
ReAct	19.36±5.07	12.12±3.59	48.60±8.54	96.47±13.49	9.18±2.54	11.40
PolySkill	13.95±3.44	8.48±3.63	36.81±7.97	76.38±15.88	10.95±2.20	9.60
Reflexion	10.06±1.93	7.89±1.84	29.28±5.85	69.56±18.70	7.88±1.80	4.00
Chain-of-Thought	19.42±5.53	13.78±2.94	54.71±12.67	103.92±14.07	10.61±2.76	12.60
Tree of Thoughts	14.07±2.59	8.56±2.60	37.14±4.88	77.18±11.37	7.72±2.89	9.40
UserSimCRS v2	14.01±3.20	8.52±1.71	36.97±5.71	76.78±19.89	10.99±3.28	10.60
CSHI	13.89±3.04	8.44±3.04	36.65±5.13	75.98±13.05	10.02±2.09	8.40
USimAgent	9.94±2.47	7.81±2.31	28.95±4.54	66.20±14.03	7.69±2.73	2.00
BASES	10.18±1.84	7.97±1.46	29.60±3.93	70.36±9.84	7.95±2.60	6.00
Agent4Rec	10.00±4.07	7.85±1.92	29.12±9.55	66.60±12.43	8.32±1.97	3.80
SimUSER	10.24±4.18	8.01±1.54	29.77±5.99	70.76±20.75	9.05±1.97	7.20
ConvSearch-R1	10.12±3.65	7.93±1.53	29.44±5.07	69.96±10.43	7.92±2.73	5.00

D Additional Aggregate Fidelity Results

The full clicked-rank comparison is reported in Table 8; Tables 9–12 provide the remaining components of aggregate fidelity. SearchAgent ranks first on click volume and abandonment, reducing the click-volume discrepancy by 0.7% on average relative to USimAgent, the second-best method on that component. It ranks second on session length and reformulation. Reflexion and Tree of Thoughts occasionally match session duration more closely, while ConvSearch-R1 is strongest on reformulation. These results delimit the source of the overall gains: SearchAgent is particularly effective at reproducing which results users click, how often they click, and whether they abandon, while specialized reasoning methods can better match individual session-level rates on some datasets.

Cross-metric interpretation. The component tables show why the aggregate PFD improvement is smaller than the clicked-rank improvement highlighted in the main text. For clicked rank, SearchAgent is first on every collection, and the reduction relative to BASES stays within 17.9–20.5%. The same cross-collection consistency appears for click volume and abandonment: the model is first on every available dataset, rather than obtaining its average rank from one unusually favorable collection. These two components are closely connected to the hierarchical policy. The macro-action layer controls whether a click or stop occurs, while the conditional selector controls where the click is placed; improvements in both components therefore indicate that the gain is not limited to redistributing a fixed number of clicks across ranks.

The remaining components expose a complementary limitation. SearchAgent is uniformly second on session length, behind Reflexion on three datasets and Tree of Thoughts on TREC14, and is uniformly second to ConvSearch-R1 on reformulation. Consequently, the model's lowest overall PFD does not mean that it dominates every behavioral marginal. Instead, it combines large and stable gains in rank choice with smaller, consistent improvements in click volume and abandonment. Residual errors remain in how long sessions continue and how often needs are reformulated. This decomposition is consistent with the main-text characterization of the average PFD gain as modest: the strongest improvements occur in the components most directly constrained by the proposed action and SERP-selection mechanisms, whereas some longer-horizon rates remain better matched by specialized reasoning policies.

Table 10: Normalized abandonment-rate discrepancy across four session datasets. ORCAS is omitted because abandonment is unavailable. Values are mean ± standard deviation over three runs (seeds 42, 123, and 456). Lower is better.

Method	TREC12	TREC13	TREC14	Yandex	ORCAS	Avg. Rank
SearchAgent (ours)	**11.21±1.93**	**7.71±2.06**	**29.16±3.60**	**65.09±10.10**	–	**1.00**
ReAct	14.35±3.30	11.80±2.76	42.06±11.46	78.49±22.14	–	11.00
PolySkill	11.85±5.01	7.99±3.25	32.02±6.13	72.27±21.06	–	8.00
Reflexion	11.40±2.32	7.79±2.43	30.16±7.46	67.89±10.41	–	3.00
Chain-of-Thought	17.73±3.31	13.08±2.66	49.02±9.81	91.72±27.25	–	13.00
Tree of Thoughts	11.96±3.83	8.07±1.46	32.35±8.06	73.07±21.52	–	10.00
UserSimCRS v2	14.41±2.99	12.34±2.05	42.22±10.91	78.89±11.59	–	12.00
CSHI	11.79±2.66	7.95±1.81	31.86±9.12	71.87±21.21	–	7.00
USimAgent	11.26±3.24	7.75±1.53	30.00±5.29	65.49±20.02	–	2.00
BASES	11.52±3.45	7.87±1.23	30.49±7.33	68.69±13.56	–	5.00
Agent4Rec	11.57±2.79	7.91±2.80	31.04±4.85	69.65±18.36	–	6.00
SimUSER	11.91±2.90	8.03±2.69	32.19±9.79	72.67±14.69	–	9.00
ConvSearch-R1	11.46±2.51	7.83±1.66	30.33±8.24	68.29±17.79	–	4.00

Table 11: Normalized session-length discrepancy across four session datasets. ORCAS is omitted because multi-query session length is unavailable. Values are mean ± standard deviation over three runs (seeds 42, 123, and 456). Lower is better.

Method	TREC12	TREC13	TREC14	Yandex	ORCAS	Avg. Rank
SearchAgent (ours)	10.75±2.19	7.46±1.43	33.38±4.19	74.14±17.44	–	2.00
ReAct	19.62±5.70	12.81±2.45	48.59±12.28	96.81±25.13	–	12.00
PolySkill	14.65±2.66	11.34±2.24	44.08±10.43	87.31±13.95	–	8.00
Reflexion	**10.69±3.87**	**6.17±1.64**	33.54±8.60	**73.74±14.07**	–	1.50
Chain-of-Thought	19.68±3.78	14.20±2.34	53.11±7.46	102.27±16.58	–	13.00
Tree of Thoughts	10.80±4.25	7.50±2.82	**10.16±7.33**	74.53±14.76	–	2.50
UserSimCRS v2	14.77±2.58	11.42±3.40	44.41±9.78	88.11±13.37	–	10.00
CSHI	14.59±2.29	10.92±2.06	43.92±6.42	75.73±18.98	–	6.75
USimAgent	10.92±3.60	7.58±1.73	33.87±5.92	75.33±12.38	–	5.00
BASES	10.98±2.33	7.62±1.62	34.03±5.45	76.13±19.35	–	6.25
Agent4Rec	14.71±2.45	11.38±1.67	44.25±8.59	87.71±10.96	–	9.00
SimUSER	14.83±2.57	11.46±1.78	44.57±5.20	88.51±11.26	–	11.00
ConvSearch-R1	10.86±1.94	7.54±1.29	33.70±4.28	74.93±19.72	–	4.00

Table 12: Normalized reformulation-rate discrepancy across four session datasets. ORCAS is omitted because query reformulation is unavailable. Values are mean ± standard deviation over three runs (seeds 42, 123, and 456). Lower is better.

Method	TREC12	TREC13	TREC14	Yandex	ORCAS	Avg. Rank
SearchAgent (ours)	10.33±2.94	6.68±2.21	23.14±4.75	66.45±11.72	–	2.00
ReAct	14.91±3.80	12.11±4.04	42.43±7.76	78.16±17.26	–	9.75
PolySkill	23.15±3.15	20.62±2.53	61.12±11.48	104.47±12.01	–	13.00
Reflexion	10.38±2.67	6.72±2.62	23.30±4.40	66.84±20.42	–	3.00
Chain-of-Thought	18.35±4.25	13.53±4.33	49.61±8.66	92.05±24.24	–	12.00
Tree of Thoughts	10.46±2.87	6.76±1.82	23.47±4.25	67.26±19.62	–	4.25
UserSimCRS v2	14.96±4.36	12.69±2.96	42.60±7.56	78.55±21.93	–	10.75
CSHI	12.44±4.00	10.70±2.71	37.93±10.00	85.65±10.73	–	9.00
USimAgent	10.51±2.21	6.81±2.54	29.29±7.68	67.65±18.21	–	5.25
BASES	10.57±3.14	6.85±2.35	29.46±8.05	68.05±11.98	–	6.25
Agent4Rec	11.87±3.61	6.72±1.47	33.37±6.43	74.27±20.06	–	6.50
SimUSER	12.50±3.86	8.13±1.97	38.09±9.08	69.19±19.19	–	8.25
ConvSearch-R1	**10.27±2.27**	**3.10±2.46**	**22.98±5.72**	**51.59±11.84**	–	**1.00**

The dataset pattern also helps rule out a single-regime explanation. The same ordering is observed on the small TREC collections, the larger structural Yandex log, and ORCAS, where only query-level click behavior is available. At the same time, missing components are not imputed: ORCAS contributes clicked-rank and

Table 13: Clicked-rank precision under matched-SERP replay on four session datasets. Results are mean ± standard deviation over three runs. Higher is better. Best results are shown in bold, and second-best results are underlined.

Method	TREC12	TREC13	TREC14	Yandex	Avg. Rank
General reasoning and agent methods					
ReAct	0.157±0.027	0.205±0.025	0.147±0.022	0.157±0.012	11.25
PolySkill	0.486±0.057	0.192±0.030	0.074±0.038	0.000±0.000	11.50
Reflexion	0.265±0.061	0.303±0.056	0.198±0.077	0.334±0.053	10.00
Chain-of-Thought	0.132±0.033	0.147±0.010	0.078±0.011	0.088±0.003	12.50
Tree of Thoughts	**0.792±0.366**	**0.889±0.184**	0.292±0.366	0.680±0.386	3.50
User modeling and search methods					
UserSimCRS v2	0.676±0.118	0.792±0.104	**0.462±0.086**	0.841±0.092	**1.75**
CSHI	0.592±0.096	0.731±0.088	0.421±0.074	0.768±0.081	4.00
USimAgent	0.427±0.047	0.552±0.075	0.293±0.079	0.379±0.030	8.00
BASES	0.523±0.059	0.535±0.163	0.272±0.079	0.219±0.083	8.75
Agent4Rec	0.466±0.074	0.672±0.055	0.305±0.102	0.480±0.060	6.75
SimUSER	0.597±0.071	0.783±0.046	0.319±0.058	0.623±0.126	4.50
ConvSearch-R1	0.657±0.037	0.789±0.074	0.410±0.152	0.867±0.134	2.75
SearchAgent (ours)	0.559±0.089	0.549±0.036	0.450±0.067	0.451±0.090	5.75

Table 14: Clicked-rank recall under matched-SERP replay on four session datasets. Results are mean ± standard deviation over three runs. Higher is better. Best results are shown in bold, and second-best results are underlined.

Method	TREC12	TREC13	TREC14	Yandex	Avg. Rank
General reasoning and agent methods					
ReAct	0.279±0.086	0.341±0.042	0.351±0.070	0.310±0.027	3.00
PolySkill	0.129±0.017	0.106±0.015	0.074±0.030	0.000±0.000	10.50
Reflexion	**0.319±0.079**	0.224±0.049	0.234±0.082	0.267±0.041	4.00
Chain-of-Thought	0.305±0.094	0.506±0.015	**0.438±0.087**	**0.393±0.010**	**1.50**
Tree of Thoughts	0.024±0.016	0.022±0.018	0.018±0.018	0.021±0.013	12.75
User modeling and search methods					
UserSimCRS v2	0.036±0.014	0.029±0.010	0.051±0.018	0.041±0.012	11.50
CSHI	0.049±0.016	0.038±0.012	0.063±0.019	0.052±0.014	10.50
USimAgent	0.267±0.043	0.215±0.034	0.265±0.110	0.326±0.018	4.00
BASES	0.186±0.036	0.071±0.020	0.117±0.045	0.029±0.018	9.63
Agent4Rec	0.212±0.060	0.136±0.018	0.253±0.084	0.105±0.028	6.50
SimUSER	0.226±0.044	0.238±0.024	0.222±0.077	0.114±0.023	5.75
ConvSearch-R1	0.186±0.036	0.132±0.021	0.160±0.076	0.064±0.023	8.13
SearchAgent (ours)	0.302±0.028	**0.583±0.011**	0.259±0.037	0.221±0.043	3.25

click-volume evidence but not abandonment, session length, or reformulation. The aggregate conclusions should therefore be read as agreement over the behaviors actually observed in each collection, not as evidence that every dataset supports the same session-level claims.

E Additional Replay Results

Tables 13 and 14 decompose replay F1. Precision varies more strongly across methods because conservative policies can obtain high scores by clicking only a small subset of ranks. Recall exhibits the opposite tendency for policies that click broadly. SearchAgent achieves the best recall on TREC13 and the second-best precision on TREC14, while remaining competitive on the other datasets. Its leading average F1 in the main text therefore reflects a favorable precision-recall balance rather than an extreme click propensity.

Precision–recall structure. The decomposed results make the operating behavior of the baselines explicit. Tree of Thoughts and UserSimCRS v2 obtain high precision on several datasets, but their recall is extremely low; for example, Tree of Thoughts reaches precision 0.792 on TREC12 while recalling only 0.024 of the observed clicked ranks. Chain-of-Thought exhibits the reverse pattern, obtaining the strongest average recall but low precision. These policies can therefore look favorable under one marginal while missing the joint requirement to reproduce both the presence and location of clicks.

SearchAgent occupies a more balanced region. On TREC13, its precision of 0.549 is below the most conservative simulators, but its recall of 0.583 is the highest in the table; this balance produces the large F1 advantage reported in the main text. TREC12 and TREC14 show the same qualitative pattern without requiring either metric

to be individually best. The Yandex exception is also diagnostic: precision remains moderate at 0.451, whereas recall falls to 0.221, below USimAgent's 0.326. Thus, the Yandex replay gap is driven more by missed observed clicks than by an excessive number of spurious ranks. This is consistent with the text-blind candidate setting and the different click regime discussed in the transfer analysis, although the replay table alone does not identify which factor is causal.

Taken together with the aggregate tables, replay separates two notions of fidelity. Low clicked-rank discrepancy shows that the simulated population has a similar rank distribution, whereas matched-SERP F1 asks whether the simulator selects the observed ranks in the same context. A method can perform well on one and poorly on the other. SearchAgent's TREC gains on both views therefore provide stronger evidence than either metric alone, while the Yandex result marks a clear boundary of that evidence.

F Additional Analyses

The analyses in this section progressively narrow the interpretation of the main results. We first equalize supervision and test whether the PFD conclusion depends on its definition or data partition. We then examine robustness across LLM backbones and collection shifts, followed by mechanism-level tests of calibration, state dependence, persona controls, and SERP perturbations. Finally, downstream transfer and feedback diagnostics assess whether the generated behavior remains useful and stable beyond the primary fidelity metrics. This ordering separates evidence for average performance from evidence about why the performance occurs and where it does not generalize.

Reading the diagnostics jointly. The additional tables should not be interpreted as interchangeable measurements of a single notion of fidelity. PFD aggregates population-level discrepancies, replay F1 matches actions in the same displayed context, and probability calibration measures confidence against logged actions. In particular, the equal- supervision Yandex rows provide a useful counterexample to a single-metric reading: the proposed model has better PFD than calibrated USimAgent (66.63 versus 68.02), but lower replay F1 (0.296 versus 0.365). On opaque-URL data, matching marginal frequencies therefore need not imply matching individual decisions. The distinction also helps interpret ORCAS, which provides a query-conditioned candidate pool but not multi-step trajectories: its PFD evidence cannot validate session-level mechanisms.

Avoiding overinterpretation. The paired randomization and bootstrap analyses compare generated and logged behaviors under the specified matched evaluation units; they do not randomize human exposure or prove that a simulator mechanism caused logged user decisions. The leave-one-TREC-out experiment evaluates transfer within a shared benchmark family, whereas Yandex has unavailable semantic fields and ORCAS uses a different interaction unit. These differences motivate reporting per-collection metrics and observable-field masks instead of claiming dataset-invariant fidelity. The evidence supports an incremental learned-behavior benefit under controlled supervision, with explicit limitations on dataset-specific significance, context-matched replay, and environment generalization.

F.1 Equal-Supervision Baselines

Table 15 applies the protocol in Appendix C.4. Equalizing supervision and tuning substantially strengthens the comparators. Relative to the strongest calibrated baseline on each dataset, the reported PFD reduction decreases from the 5.2% native-regime comparison to 1.3–3.7% (2.4% macro average). The pooled paired interval is [0.8%, 4.0%], while the TREC13 and Yandex dataset-level intervals include zero. SearchAgent remains strongest in PFD and TREC replay, but calibrated USimAgent remains better on Yandex replay. This control supports a narrower conclusion: the proposed decomposition adds an incremental benefit beyond access to supervision, rather than supervision alone explaining the entire result.

What remains after supervision is equalized. The strongest equally supervised competitor is BASES+calibration for PFD on all five datasets. The remaining reductions are 2.0%, 2.5%, 2.7%, 1.3%, and 3.7% on TREC12, TREC13, TREC14, Yandex, and ORCAS, respectively. Their consistency in sign explains the positive pooled interval, but their small magnitude also prevents attributing the native-regime gap entirely to the proposed architecture. The TREC replay advantage is similarly incremental: 0.428 versus 0.419 for BASES+calibration. In contrast, Yandex replay favors USimAgent+calibration by a substantial margin (0.365 versus 0.296). The controlled comparison therefore supports two distinct statements: the gate–residual–selector decomposition retains a measurable average advantage under matched supervision, and that advantage is neither large nor universal across replay settings.

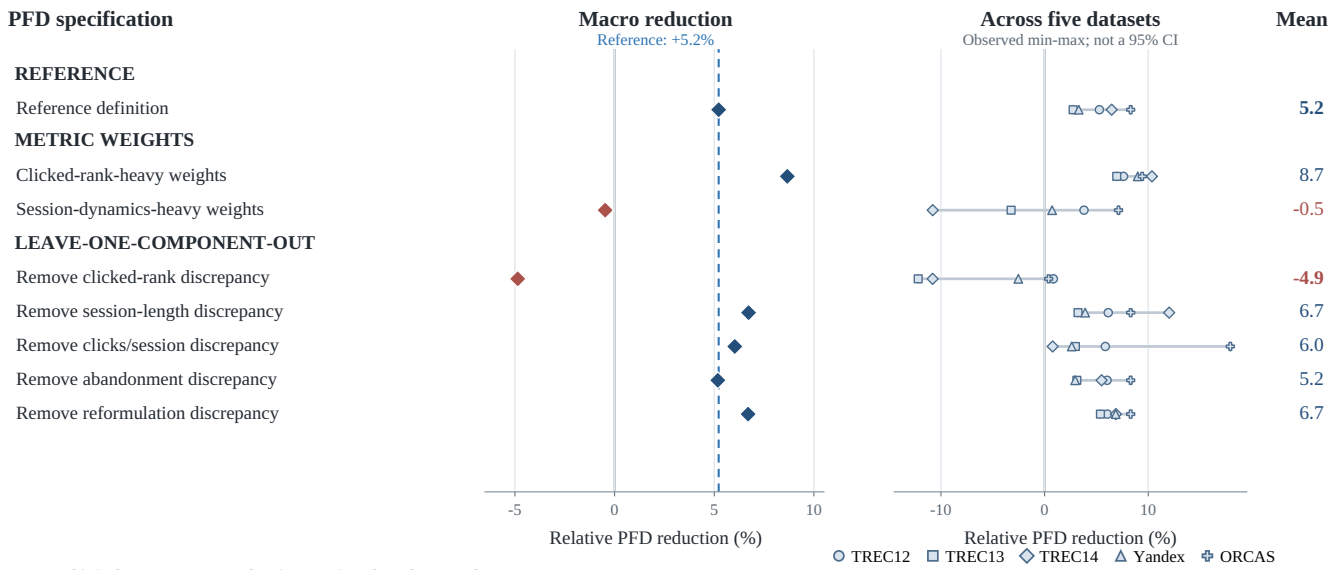
F.2 PFD Definition and Partition Sensitivity

To test the dependence of aggregate fidelity on the PFD definition, we compare equal weighting with clicked-rank-heavy $(0.35, 0.15, 0.25, 0.15, 0.10)$ and session-dynamics-heavy $(0.10, 0.30, 0.15, 0.25, 0.20)$ weights, following the component order clicked rank, session length, click volume, abandonment, and reformulation. Weights are renormalized over observable metrics in each dataset; ORCAS contains only clicked rank and click volume. We additionally remove each metric in turn and renormalize the remaining weights. For each scenario, we recompute the macro relative reduction against the lowest-PFD baseline selected separately for each dataset. To summarize the matched-supervision results without introducing an independent aggregation procedure, we also report datasetwise PFD ranks and pairwise win rates computed directly from Table 15. These descriptive statistics should not be interpreted as raw-component aggregation robustness tests.

Figure 1 reports macro relative PFD reductions of 5.22% (equal weights), 8.67% (clicked-rank-heavy), and −0.47% (session-dynamics-heavy). Omitting clicked rank changes the reduction to −4.86%; omitting session length, click volume, abandonment, or reformulation instead gives 6.72%, 6.03%, 5.18%, and 6.70%, respectively. Baselines are reselected for each dataset and definition, so negative reductions signify higher PFD than the strongest comparator. Dataset markers reveal substantial heterogeneity, particularly without clicked rank. These are deterministic recalculations from rounded published component means, not per-seed re-estimates; the displayed ranges do not represent bootstrap uncertainty. Thus the aggregate advantage depends on emphasizing

Table 15: Results under equal log supervision, comparable trainable capacity, and the same 36-trial validation budget. PFD is lower-is-better; replay F1 is higher-is-better. Best and second-best entries are bold and underlined, respectively.

Method	TREC12	TREC13	TREC14	Yandex	ORCAS	TREC F1	Yandex F1
Supervised behavioral models							
PBM + Markov session	12.08	8.21	34.65	75.92	6.44	0.354	0.242
DBN + session controller	11.62	7.84	32.18	73.60	6.21	0.369	0.251
Neural click–session model	10.47	7.04	29.12	69.42	5.90	0.406	0.273
Behavior cloning MLP	10.56	7.08	29.41	70.15	5.88	0.402	0.268
Raw LLM prior + plain MLP	10.24	6.93	28.32	68.74	5.69	0.414	0.281
Equally supervised agent baselines							
USimAgent + calibration	10.28	6.97	28.56	68.02	5.74	0.416	**0.365**
BASES + calibration	<u>10.05</u>	<u>6.88</u>	<u>27.92</u>	<u>67.51</u>	<u>5.62</u>	<u>0.419</u>	<u>0.327</u>
SearchAgent (ours)	**9.85**	**6.71**	**27.17**	**66.63**	**5.41**	**0.428**	0.296



Points: published component means; baseline re-selected per dataset and scenario.

Figure 1: Deterministic PFD sensitivity under alternative metric weights and leave-one-component-out definitions. Diamonds are macro averages across five datasets; small markers show individual datasets, with observed minimum–maximum spans (not confidence intervals). Each dataset and scenario uses its own lowest-PFD baseline. Positive reductions favor SearchAgent; negative reductions favor the baseline. All values are reconstructed from rounded component means in Tables 8–12.

Table 16: Equal-supervision PFD ranks across datasets (lower is better). Mean ranks and pairwise win rates are determined by the same five PFD values; each method faces 35 dataset–opponent comparisons.

Method	TREC12	TREC13	TREC14	Yandex	ORCAS	Mean rank ↓	Win rate ↑
SearchAgent	1	1	1	1	1	**1.00**	**100.0%**
BASES + calibration	2	2	2	2	2	<u>2.00</u>	<u>85.7%</u>
Raw LLM prior + plain MLP	3	3	3	4	3	3.20	68.6%
USimAgent + calibration	4	4	4	3	4	3.80	60.0%
Neural click–session model	5	5	5	5	6	5.20	40.0%
Behavior cloning MLP	6	6	6	6	5	5.80	31.4%
DBN + session controller	7	7	7	7	7	7.00	14.3%
PBM + Markov session	8	8	8	8	8	8.00	0.0%

clicked-rank fidelity and does not hold under every reasonable metric specification.

Table 16 re-expresses the PFD values in Table 15; its win rate is determined by mean rank and cannot independently test raw-metric scaling. Such a test requires equal-supervision component-level discrepancies for all methods.

To assess partition sensitivity under the native-regime protocol used for the main paper's primary PFD comparison, we repeat grouping, model selection, training, and evaluation across ten independently hashed split seeds while preserving the same train/validation/test proportions and leakage constraints. Hyperparameters are reselected inside each validation partition; the test set is untouched until selection is complete.

Under the native-regime protocol, the ten-split macro reduction averages 4.78%, compared with 5.2% on the original fixed split reported in the main paper. This robustness analysis uses the primary

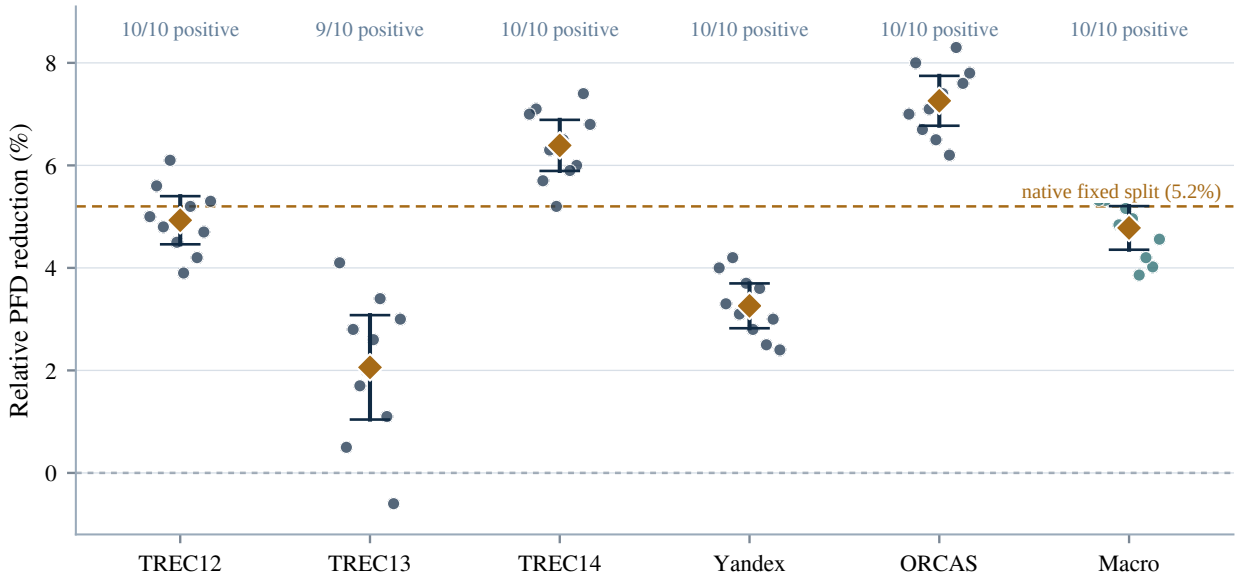


Figure 2: Partition sensitivity of the primary PFD comparison across ten independently generated, leakage-safe splits under the native-regime protocol. Circles show split-specific relative reductions; diamonds and bars show means and 95% t intervals. The dashed line marks the native-regime fixed-split result (5.2%) from the main paper; the equal-supervision fixed-split result (2.4%) is a separate control.

Table 17: Relative PFD reductions across ten leakage-safe splits under the main paper's native-regime comparison protocol, distinct from the equal-supervision control in Table 15. Entries give mean, standard deviation, range, and positive-split count.

Dataset	Mean ± SD	Range	Positive
TREC12	4.93 ± 0.66%	3.9–6.1%	10/10
TREC13	2.06 ± 1.42%	−0.6–4.1%	9/10
TREC14	6.39 ± 0.70%	5.2–7.4%	10/10
Yandex	3.26 ± 0.61%	2.4–4.2%	10/10
ORCAS	7.26 ± 0.68%	6.2–8.3%	10/10
Macro	**4.78 ± 0.59%**	**3.86–5.56%**	**10/10**

comparison's training conditions, rather than the equal-supervision control in Table 15, which yields 2.4% on its fixed split. The two protocols employ different baseline training conditions and should not be pooled. TREC13 shows the greatest split sensitivity, including one negative split. All ten macro reductions remain positive, but this does not imply uniform dataset-level dominance.

The PFD ranks in Table 16 are consistent with the equal-supervision comparison, not an independent raw-aggregation robustness test. All ten native-regime partitions yield positive macro reductions (Table 17), whereas excluding clicked-rank fidelity reverses the macro difference. The gain is therefore sensitive to the selected PFD definition.

F.3 Cross-Backbone Gate and Residual Robustness

We keep prompts, field masks, train/validation/test partitions, learned-head architecture, optimization budget, and random seeds fixed while replacing only the provider-hosted semantic-prior model. The calibrator is refitted for each backbone from training data; no test-time rescaling is allowed. Raw prior, reliability-gated prior, and full gate-plus-residual variants isolate whether the learned correction remains useful when the upstream probability geometry changes. The requested July 2026 API identifiers are deepseek-v4-pro, gpt-5.6-sol, and claude-sonnet-5; the returned provider model IDs are stored in the run manifest rather than assumed from aliases.

Across all three backbones, the Gate improves Replay F1 and PFD over the raw prior. Action NLL decreases with gating for GPT-5.6 Sol and Claude Sonnet 5, but increases from 1.246 to 1.308 for DeepSeek-V4-Pro. Gate+Residual further improves Replay F1, PFD, and Action NLL over the gated variant for every backbone, with low OrthErr. Thus, the gate's behavioral-fidelity gains do not necessarily imply improved action likelihood. Absolute cross-provider rankings remain secondary because probability elicitation may differ across APIs.

Backbone-relative effects. The three policy variants are consistently ordered by Replay F1 and PFD within each backbone, but not by Action NLL. Moving from the raw prior to the full model raises Replay F1 from 0.310 to 0.428 for DeepSeek-V4-Pro, from 0.339 to 0.441 for GPT-5.6 Sol, and from 0.328 to 0.435 for Claude Sonnet 5. PFD falls by roughly 35–38% in the same comparisons. The adaptive gate provides the larger first improvement in Replay F1 and PFD, while the residual adds further gains in both metrics and lowers Action NLL below the raw prior for every backbone. Because the learned heads are refitted separately, this test does not show that one fixed calibrator transfers unchanged across providers; it shows that the decomposition remains useful when trained against different elicited semantic priors. The consistently small OrthErr further

Table 18: Cross-backbone robustness, macro-averaged over TREC12–14. Values are reported as mean and standard deviation over three runs. Higher Replay F1 is better; lower PFD, Action NLL, and OrthErr are better. The evaluation protocol is shared across backbones.

Semantic-prior model	Policy variant	Replay F1 ↑	PFD ↓	Action NLL ↓	OrthErr ↓
DeepSeek-V4-Pro	Raw prior	0.310±0.018	23.43±3.82	1.246±0.031	–
DeepSeek-V4-Pro	+ Gate	0.381±0.030	17.12±3.34	1.308±0.034	–
DeepSeek-V4-Pro	+ Gate + Residual	0.428±0.018	14.58±2.90	0.998±0.019	0.003±0.001
GPT-5.6 Sol	Raw prior	0.339±0.029	21.76±4.10	1.421±0.037	–
GPT-5.6 Sol	+ Gate	0.402±0.027	16.45±3.07	1.276±0.031	–
GPT-5.6 Sol	+ Gate + Residual	0.441±0.026	14.09±2.74	1.203±0.027	0.005±0.002
Claude Sonnet 5	Raw prior	0.328±0.031	22.31±4.25	1.447±0.039	–
Claude Sonnet 5	+ Gate	0.394±0.028	16.81±3.16	1.291±0.032	–
Claude Sonnet 5	+ Gate + Residual	0.435±0.027	14.36±2.80	1.211±0.028	0.006±0.002

Table 19: Cross-collection transfer. TREC columns are macro-averaged over TREC12–14. Values are mean ± standard deviation over three runs; higher F1 and lower PFD are better.

Training protocol	TREC target		Yandex target		ORCAS target	
	F1 ↑	PFD ↓	F1 ↑	PFD ↓	F1 ↑	PFD ↓
Raw structured LLM prior	0.310±0.018	23.43±3.82	0.214±0.024	94.80±17.60	0.521±0.018	9.42±2.76
Single-source: TREC	–	–	0.246±0.026	82.15±15.20	0.566±0.017	7.31±2.25
Single-source: Yandex	0.352±0.031	19.06±3.72	–	–	0.548±0.019	8.04±2.41
Single-source: ORCAS	0.365±0.030	18.22±3.51	0.238±0.025	85.60±15.90	–	–
Leave-target-out pooled	0.391±0.029	16.74±3.18	0.261±0.027	78.36±14.70	0.589±0.016	6.62±2.06
In-domain upper bound	**0.428±0.018**	**14.58±2.90**	**0.296±0.056**	**66.63±14.07**	**0.612±0.015**	**5.41±2.12**

indicates that this gain is not obtained by allowing the residual to collapse back into a second confidence-scaling mechanism.

F.4 Cross-Collection Transfer over TREC, Yandex, and ORCAS

The transfer unit is the learned behavioral layer; LLM prompts and dataset field masks remain target specific. "Leave-target-out pooled" trains the gate, residual, and click selector on the other two collection families and selects hyperparameters without the target test set. On Yandex, $m^{\text{text}} = 0$ removes semantic candidate information. ORCAS replay F1 is computed over the clicked candidate versus non-click outcome in its fixed query-conditioned pool and is not interpreted as a session metric.

Transfer improves F1 and PFD over the raw prior for all three targets, and leave-target-out pooling is the strongest transfer setting. The TREC reference rows match Table 21, including their standard deviations. Because targets differ in metric scale and available fields, raw PFD gaps should not be compared across collections; these aggregate metrics also do not isolate which behavioral mechanisms transfer.

Transfer asymmetry. Leave-target-out pooling closes a substantial portion of the gap between the raw prior and in-domain training, but the recovery is not uniform. On the TREC target, pooled transfer reaches F1 0.391 and PFD 16.74, compared with the in-domain values 0.428 and 14.58. ORCAS shows a similar pattern, with pooled transfer reaching F1 0.589 and PFD 6.62. Yandex recovers the smallest fraction of the raw-prior–in-domain gain: 57.3%/58.4% (F1/PFD), compared with 68.6%/75.6% for TREC and 74.7%/69.8% for ORCAS. Pooled Yandex transfer reaches F1 0.261 and PFD 78.36 versus the in-domain values 0.296 and 66.63. These descriptive differences are consistent with missing candidate text and different session structure, but do not establish either factor as the sole cause.

Single-source transfer further suggests that pooling is not merely averaging incompatible models. For each target, the pooled model outperforms either single-source alternative on both F1 and PFD. The shared layer therefore captures behavioral regularities that recur across collection families, while the remaining in-domain advantage reflects target-specific calibration. Since prompts and masks remain target specific, the experiment isolates transfer of the learned behavioral layer rather than claiming a fully dataset-agnostic end-to-end simulator.

F.5 Cross-Dataset Generalization Details

Table 20 provides the complete leave-one-TREC-dataset-out results summarized in the main paper.

Per-target recovery. The source-only results are notably uniform across the three held-out TREC datasets. Relative to the gap between the raw prior and in-domain training, source-only transfer recovers approximately 67–70% of the Replay-F1 gain and 68–71% of the PFD improvement on each target. The macro recovery reported in the main text is therefore not produced by one easy transfer direction. TREC14 retains the largest absolute gap because its raw-prior PFD is much higher, but its proportional recovery is comparable to TREC12 and TREC13.

Table 20: Leave-one-TREC-dataset-out transfer. Source-only models are trained on the other two TREC datasets. The target-calibrated variant uses a disjoint 10% calibration partition and does not access test sessions. Values are mean ± standard deviation over three runs; macro averages are computed across TREC12–14.

Training Protocol	TREC12		TREC13		TREC14		Macro Avg.	
	F1 ↑	PFD ↓	F1 ↑	PFD ↓	F1 ↑	PFD ↓	F1 ↑	PFD ↓
Raw LLM prior	0.286±0.031	16.30±3.20	0.401±0.026	11.20±2.10	0.242±0.041	42.80±7.90	0.310	23.43
Source-only transfer	0.356±0.035	11.75±2.62	0.515±0.023	8.15±1.80	0.302±0.041	31.90±6.12	0.391	17.27
Source + 10% target calibration	0.374±0.036	10.65±2.45	0.540±0.022	7.35±1.72	0.315±0.043	29.40±5.58	0.410	15.80
In-domain training	**0.390±0.037**	**9.85±2.34**	**0.565±0.020**	**6.71±1.66**	**0.329±0.045**	**27.17±5.05**	**0.428**	**14.58**

Table 21: Probability calibration, orthogonality, and behavioral fidelity, macro-averaged over TREC12–14. The no-projection variant retains the adaptive gate but removes prior-orthogonal projection; the fixed-gate variant retains the orthogonal residual but uses context-independent prior reliability. Values are mean ± standard deviation over three runs.

Variant	NLL ↓	Brier ↓	ECE ↓	OrthErr ↓	Replay F1 ↑	PFD ↓
Raw LLM prior	1.246±0.031	0.512±0.014	0.173±0.018	–	0.310±0.018	23.43±3.82
Temperature scaling	1.173±0.027	0.486±0.013	0.106±0.014	–	0.325±0.017	21.91±3.51
Adaptive gate + residual w/o projection	1.022±0.021	0.421±0.010	0.057±0.009	0.086±0.012	0.411±0.016	15.84±2.96
Orthogonal residual + fixed gate	1.052±0.023	0.434±0.011	0.071±0.011	0.003±0.001	0.397±0.017	17.11±3.10
Full model	**0.998±0.019**	**0.407±0.009**	**0.041±0.007**	**0.003±0.001**	**0.428±0.018**	**14.58±2.90**

The disjoint 10% target calibration split reduces the remaining gap on every dataset and on both metrics. Its recovery is approximately 84–85% for Replay F1 and 86–88% for PFD, while still leaving the test sessions untouched. Limited target evidence therefore appears to correct collection-specific scale and action-frequency differences rather than replacing the transferable structure. The residual gap to in-domain training also cautions that the behavioral layer is only partially invariant: full target supervision remains useful, especially for the higher-discrepancy TREC14 setting.

F.6 Calibration and Orthogonality Details

Table 21 reports the complete probability quality, orthogonality, and behavioral-fidelity results summarized in the main paper.

Metric-by-metric interpretation. Temperature scaling primarily improves probability calibration: ECE falls from 0.173 to 0.106, but Replay F1 rises only from 0.310 to 0.325 and PFD falls only from 23.43 to 21.91. This separates confidence correction from behavioral correction. Adding an unconstrained residual yields a much larger fidelity gain, but its OrthErr of 0.086 shows that part of the residual remains aligned with the centered log-prior. Orthogonal projection reduces this overlap to 0.003. Comparing the no-projection variant with the full model then shows that identifiability is obtained without sacrificing performance: Replay F1 increases from 0.411 to 0.428 and PFD decreases from 15.84 to 14.58.

The fixed-gate comparison isolates the remaining role of context-dependent reliability. With an orthogonal residual already present, replacing the fixed weight by the adaptive gate improves NLL, Brier score, ECE, Replay F1, and PFD simultaneously. Thus, the gate is not merely another source of capacity; it adjusts how much semantic evidence is retained across states, while the projected residual represents behavioral information in a direction that the gate cannot reproduce. These comparisons support the division of labor stated in the main text rather than only the superiority of the final row.

F.7 Closed-Loop State Ablation

To evaluate the sequential state representation independently of the core action-policy ablations, we remove the persona, recent session memory, cognitive-intent state, or post-click feedback while retaining the remaining modules and retraining the calibrator under the same protocol.

The ablations reveal a clear division of labor among state variables. Removing session memory causes the largest replay-F1 drop (from 0.428 to 0.382), confirming that matched-SERP decisions depend strongly on recent interaction history. Removing post-click feedback produces the largest degradation in PFD, session-length JSD, reformulation gap, and abandonment agreement, which shows that utility and dwell feedback are essential for closing the session-level loop. Persona and cognitive-intent state have smaller effects, but every metric moves in the unfavorable direction when either module is removed; their value therefore lies in consistent personalization and state refinement rather than a single dominant aggregate effect.

Figure 4 standardizes these heterogeneous changes by the full-model standard deviation and summarizes their across-metric dispersion.

F.8 Persona Controllability

Table 23 reports paired interventions in which the query, SERP, model, and random seed are held fixed. Patience is the most stable trait: increasing it produces more clicks, longer sessions, and lower abandonment on all three TREC datasets. Expert personas consistently make fewer clicks, whereas click-depth effects vary across datasets. Decision style also affects examination depth on two datasets.

Interpretation and limits. The stable signs for patience and click count under expertise indicate that the controls have predictable behavioral effects when all other inputs are held fixed. Their magnitudes, however, differ substantially across datasets; for

Table 22: Closed-loop state ablations, macro-averaged over TREC12–14. Session JSD compares session-length distributions, Reform. Gap is the absolute reformulation-rate difference, and Abandon Agr. is replay abandonment agreement. Values are mean ± standard deviation over three runs.

Variant	Replay F1 ↑	PFD ↓	Session JSD ↓	Reform. Gap ↓	Abandon Agr. ↑
Full model	**0.428±0.018**	**14.58±2.90**	**0.118±0.016**	**0.071±0.011**	**0.689±0.025**
w/o Persona	0.401±0.020	16.21±3.05	0.136±0.019	0.090±0.014	0.653±0.031
w/o Session Memory	0.382±0.022	18.74±3.42	0.177±0.023	0.129±0.018	0.625±0.035
w/o Cognitive/Intent State	0.395±0.021	17.46±3.20	0.158±0.021	0.113±0.016	0.641±0.033
w/o Post-Click Feedback	0.388±0.023	19.08±3.60	0.187±0.025	0.138±0.019	0.612±0.038

Table 23: Controlled persona interventions with all non-intervened inputs fixed. Values denote changes from the first trait setting to the second. Higher click ranks indicate deeper examination.

Trait Change	Behavior	Direction	TREC12	TREC13	TREC14	Consistent
Patience: low → high	Clicks	↑	+0.10	+1.90	+0.13	**3/3**
Patience: low → high	Session length	↑	+0.50	+0.37	+0.53	**3/3**
Patience: low → high	Abandonment	↓	−0.08	−0.42	−0.33	**3/3**
Expertise: novice → expert	Clicks	↓	−1.48	−5.47	−1.72	**3/3**
Expertise: novice → expert	Click rank	↓	−0.62	+0.12	+3.16	1/3
Style: impulsive → deliberate	Click rank	↑	−1.34	+0.24	+0.65	2/3

Table 24: Persona initialization inputs, training targets, test-time availability, and prohibited information. Targets are behavioral proxies on training folds, not demographic labels.

Coordinate	Observable predictor inputs	Training-fold proxy target	Test-time value	Explicitly prohibited
Expertise z^{exp}	Initial-query length, mean collection IDF, entity/number/question indicators, initial query–SERP lexical overlap, and score dispersion.	Cross-fitted composite of query specificity and inverse reformulation burden, computed only for training sessions.	Prediction from the initial query and first SERP; clipped to $[0, 1]$.	Current-session clicks, later queries, dwell, utility, final success, test labels.
Decision style z^{style}	The same pre-action features plus initial SERP result-count and score-concentration features.	Training-fold early-action proxy based on normalized pre-first-click examination depth.	Cross-fitted prediction before the first action.	First clicked rank or any action from the held-out session.
Patience z^{pat}	Initial query and SERP features only; no user-history aggregate is used in the public benchmarks.	Training-fold continuation proxy combining normalized session length and continuation after an unrewarding interaction.	Cross-fitted prediction before the first action.	Held-out session length, abandonment, reformulation count, or future utility.

example, the click increase from greater patience is much larger on TREC13 than on TREC12 or TREC14. The controls should therefore be understood as directionally meaningful interventions, not as globally calibrated human-trait scales. The inconsistent click-rank response to expertise and the two-of-three result for decision style provide an important counterexample: semantic trait labels do not automatically imply the same examination behavior in every collection. This mixed pattern supports controllability for selected outcomes while avoiding a stronger claim that the persona coordinates recover latent psychological attributes.

F.9 Observable-Only Persona Initialization Protocol

The three persona coordinates are simulator controls, not directly observed human attributes. To avoid using behavior that would be unavailable when a new simulated session begins, the contextual variant is implemented as a cross-fitted, observable-only initializer. Its input is frozen immediately after the initial query and first SERP have been constructed and before any click, skip, reformulation, dwell time, satisfaction score, or stopping outcome is revealed. Table 24 gives the complete mapping.

For each outer train/validation/test partition, the initializer is trained only on the outer training partition. Five inner folds generate out-of-fold persona values for calibrator training; one final initializer fitted on the complete outer training partition predicts validation and test personas. Normalization, IDF statistics, target quantiles, and clipping thresholds are fitted inside the same training boundary. No persistent identity is required, and no persona prediction is updated from the held-out session after step zero. This protocol explains why initialization can improve matched-SERP replay while occasionally worsening population PFD: it changes conditional heterogeneity without forcing the marginal persona mixture to match the target collection.

F.10 Observable-Only Contextual Persona Initialization

Table 25 compares observable-only contextual persona initialization with a fixed neutral persona. Contextual initialization improves clicked-rank F1 on all three datasets, yielding a 14.1% macro-average gain, and increases abandonment agreement by 6.0%. Its effect on aggregate fidelity is more heterogeneous: PFD improves on TREC12 and TREC13 but worsens on TREC14, leaving the macro average nearly unchanged. This contrast suggests that persona initialization primarily strengthens context-conditioned action alignment, whereas population-level calibration remains sensitive to the dataset's session composition. Because training is repeated only three times, we emphasize effect magnitude and variability; statistical uncertainty is estimated by pairing methods on the same

Table 25: Observable-only contextual versus fixed-neutral persona initialization. Results are reported over three runs. For higher-is-better metrics, Δ is Contextual minus Fixed; for PFD, Δ is Fixed minus Contextual, so positive values consistently indicate improvement.

Dataset	Metric	Fixed	Contextual	Δ	Gain
TREC12	Click F1 ↑	0.270±0.044	**0.390±0.037**	+0.1200	+44.4%
TREC12	Abandon Agr. ↑	0.575±0.099	**0.625±0.118**	+0.0500	+8.7%
TREC12	PFD ↓	13.75±2.85	**9.85±2.34**	+3.90	+28.4%
TREC13	Click F1 ↑	0.550±0.030	**0.565±0.020**	+0.0150	+2.7%
TREC13	Abandon Agr. ↑	0.883±0.024	0.883±0.024	0.0000	0.0%
TREC13	PFD ↓	8.40±1.92	**6.71±1.66**	+1.69	+20.1%
TREC14	Click F1 ↑	0.305±0.051	**0.329±0.045**	+0.0240	+7.9%
TREC14	Abandon Agr. ↑	0.492±0.084	**0.558±0.073**	+0.0660	+13.4%
TREC14	PFD ↓	**21.60±4.72**	27.17±5.05	−5.57	−25.8%
Macro Avg.	**Click F1 ↑**	0.3750	**0.4280**	+0.0530	**+14.1%**
Macro Avg.	**Abandon Agr. ↑**	0.6500	**0.6887**	+0.0387	**+6.0%**
Macro Avg.	**PFD ↓**	14.5833	**14.5767**	+0.0066	**+0.05%**

held-out sessions and bootstrap indices rather than treating the three seed means as independent samples (Appendix C).

Why replay and PFD can move differently. The table makes the distinction concrete. Contextual initialization produces its largest replay gain on TREC12 and improves abandonment agreement on TREC12 and TREC14, but TREC13 abandonment is unchanged at an already high level. For PFD, the gains of 3.90 on TREC12 and 1.69 on TREC13 are almost exactly offset by the 5.57 degradation on TREC14, leaving the macro average unchanged. This cancellation is compatible with improved conditional decisions: assigning different pre-action personas can help select the observed action for a given SERP while shifting the overall mixture of simulated session types away from the collection-level marginal. The initializer is therefore useful for personalized replay, but it should not be treated as a substitute for population-level calibration.

F.11 Downstream Utility

Table 26 tests whether generated clicks can train models that transfer to real behavior. Models trained on generated data approach the real-data reference under both position-only and contextual click prediction. Adding session context raises Gen.→Real AUC from 0.7124 to 0.7474, indicating that the transferred signal extends beyond rank alone. The generated and real rank-CTR curves are also strongly correlated ($\rho = 0.862$), providing complementary evidence of position-bias alignment.

Transfer gap and interpretation. Training on generated clicks leaves a small but visible gap to training on real clicks. For the position-only model, Gen.→Real AUC is 0.7124 versus 0.7260 for Real→Real; for the contextual model, the corresponding values are 0.7474 and 0.7642. At the same time, adding context improves both training regimes, and the generated-data model preserves most of that gain. This indicates that the generated sessions preserve dependencies involving query position, prior clicks, and session length, rather than reproducing only the marginal rank curve.

Accuracy is nearly unchanged because it is dominated by the majority non-click outcome, whereas AUC and log likelihood expose

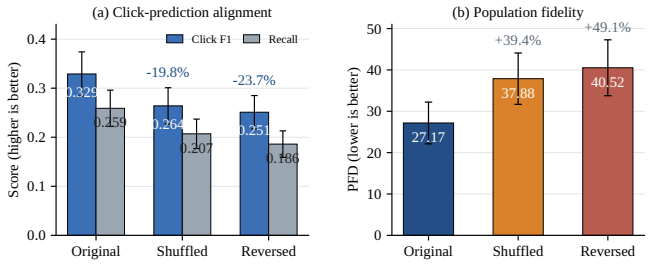


Figure 3: Sensitivity to SERP ranking perturbations on TREC 2014. Bars show mean replay F1, replay recall, and PFD over seeds 42, 123, and 456. Shuffled and reversed rankings reduce replay alignment and increase population-level discrepancy; PFD uses the same scale as Table 3.

the remaining distributional difference. The rank-CTR correlation of 0.862 is therefore complementary, not sufficient on its own: it confirms transfer of position bias, while the contextual AUC tests whether generated sessions support prediction beyond rank. These results establish downstream utility for the tested click models, but do not imply that generated data is interchangeable with real logs for arbitrary retrieval or policy-learning tasks.

F.12 SERP Ranking Sensitivity

Figure 3 evaluates SearchAgent under original, shuffled, and reversed rankings on TREC 2014. Click F1 decreases from 0.329 to 0.264 and 0.251, and replay recall decreases from 0.259 to 0.207 and 0.186. Using the same PFD scale as the full-model result in Table 3, PFD increases from 27.17 to 37.88 and 40.52. The consistent degradation under both perturbations shows that the policy responds to ranked content rather than replaying a fixed click distribution.

Perturbation severity. Shuffling reduces Replay F1 by 19.8% and raises PFD by 39.4%, while reversal reduces F1 by 23.7% and raises PFD by 49.1%. The stronger effect of reversal is consistent with the monotone examination component: moving highly ranked items systematically to low positions is more disruptive than a random permutation on average. Because both perturbations also

Table 26: Downstream transfer of generated clicks. AUC is reported as mean ± standard deviation over three data splits generated with seeds 42, 123, and 456. Blue rows denote models trained on generated clicks. The contextual model uses rank, query position, prior clicks, and session length. Spearman ρ compares per-rank CTR curves.

Model	Train → Test	AUC ↑	LogLL ↑	Acc. ↑	Spearman ρ
Click-model transfer					
Position-only	Real → Real	0.7260±0.0052	−0.1929	0.9461	–
Position-only	**Gen. → Real**	**0.7124±0.0038**	**−0.2090**	**0.9461**	–
Position-only	Gen. → Gen.	0.8323±0.0240	−0.2274	0.9135	–
Position-only	Real → Gen.	0.8331±0.0210	−0.2458	0.9135	–
Contextual LR	Real → Real	0.7642±0.0059	−0.1863	0.9459	–
Contextual LR	**Gen. → Real**	**0.7474±0.0015**	**−0.2004**	**0.9421**	–
Position-bias transfer					
Rank CTR	Gen. vs. Real	–	–	–	**0.862**
Rank CTR	Gen. vs. Joachims	–	–	–	0.862
Rank CTR	Real vs. Joachims	–	–	–	1.000

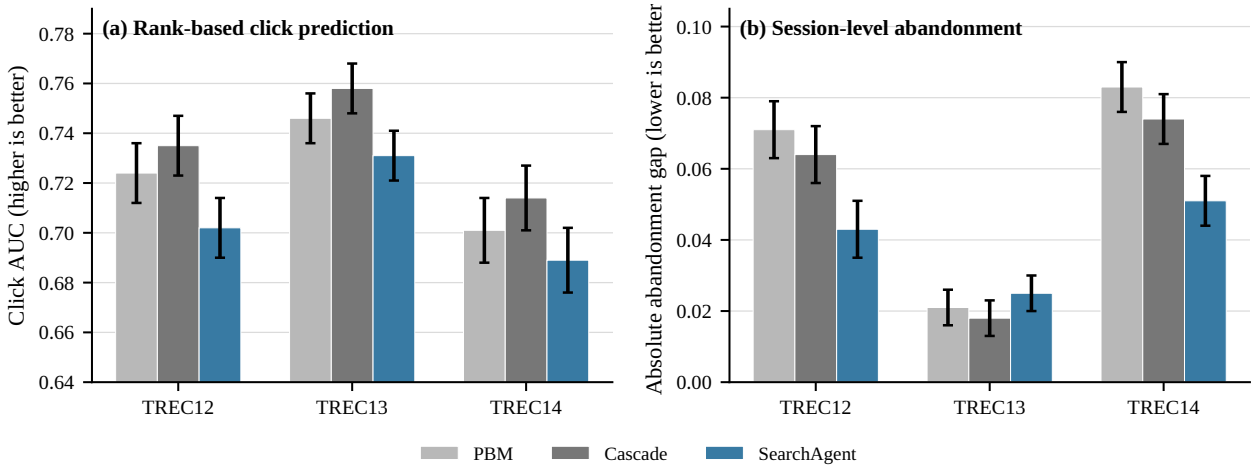


Figure 4: Comparison with classical click models on TREC12–14. Left: rank-based click AUC (higher is better). Right: absolute abandonment-rate gap (lower is better). Bars show the reported means and error bars from the comparison, without inferring an unreported significance test. PBM and Cascade achieve higher click AUC, while SearchAgent has a smaller abandonment gap on TREC12 and TREC14.

alter which content is exposed at each rank, the experiment should be interpreted as an end-to-end ranking-sensitivity test rather than a direct estimate of the examination parameter alone. Its main implication is that the simulator does not preserve its original decisions when the ranked context is changed.

F.13 Comparison with Classical Click Models

An additional comparison with classical click models separates local click prediction from session-level behavioral alignment. PBM and Cascade obtain higher click AUC on all three TREC datasets, as expected from models optimized directly for rank-conditioned clicking. The difference is modest, ranging from roughly two to three AUC points. In contrast, SearchAgent yields the smallest abandonment gap on TREC12 and TREC14, while Cascade is slightly better on the low-abandonment TREC13 set. These results indicate that the proposed simulator trades a small amount of specialized click-prediction accuracy for improved coverage of full-session behavior, including reformulation, stopping, post-click feedback, and dwell-time dynamics. Figure 4 shows the underlying comparison; the higher local click AUC of the classical models should not be interpreted as superior full-session fidelity.

F.14 Position-Bias Feedback

Figure 5 examines repeated reintroduction of the generated normalized rank-CTR profile as the position prior. In matched-SERP replay, Gini changes from 0.731 to 0.730 and then 0.716; under free-running generation, it decreases from 0.781 to 0.767 and 0.757. Thus, the tested feedback procedure provides no evidence of increasing click concentration. This diagnostic does not by itself identify which component causes the change; a causal attribution would require a separate controlled intervention.

The decrease is small but consistent across the two generation modes: from the first to the third round, Gini falls by about 2.1% in replay and 3.1% in free-running sessions. This argues against a short-horizon positive-feedback failure in which generated top-rank clicks are repeatedly amplified into a more concentrated prior.

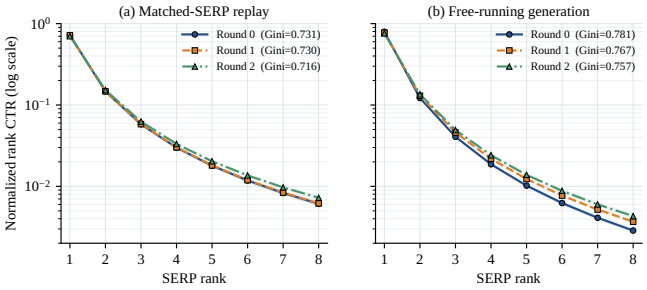


Figure 5: Position-bias feedback over three rounds. The Gini coefficient of the mean rank-CTR profile remains stable or decreases under both matched-SERP replay and free-running generation after the learned position prior is reintroduced.

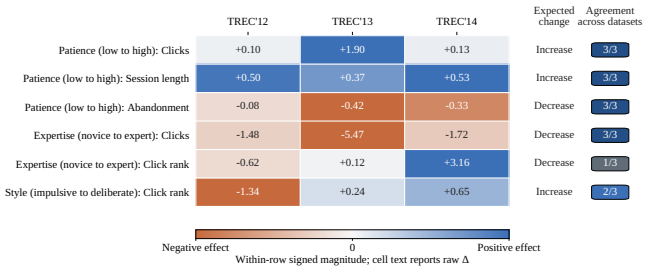


Figure 6: Controlled persona interventions across the TREC datasets. Cell text reports the raw behavioral change from the first trait setting to the second; color encodes signed magnitude normalized within each row. Numerical values correspond to Table 23.

The diagnostic remains limited to three rounds and to the mean rank-CTR profile; it does not exclude concentration within particular queries, personas, or longer feedback cycles.

F.15 Persona-Behavior Intervention Consistency

Figure 6 visualizes the controlled persona interventions reported in Table 23. Increasing patience produces more clicks, longer sessions, and lower abandonment on all three TREC datasets, while expert personas consistently make fewer clicks. In contrast, the effect of expertise on click depth varies across datasets. A deliberate decision style leads to deeper clicks on two of the three datasets. The figure summarizes controlled directional evidence and is not a causal graph.

G Exact LLM Prompts and Decoding Controls

This appendix specifies the complete prompted interfaces used by SearchAgent. The primary requested model is deepseek-v4-pro under provider-hostedstructured-outputAPI; cross-backbone requests use gpt-5.6-sol and claude-sonnet-5. Because all three are provider hosted, reproducibility is anchored by requested and returned model identifiers, request metadata, canonical prompt bytes, structured-output schema, and response hashes rather than local weight or tokenizer hashes. No prompt contains evaluation labels, aggregate click statistics, future session outcomes, environment-side relevance scores, or instructions to use tools or web search.

G.1 Structured Probability Elicitation

For each finite-label interface, the API is instructed to return exactly one JSON object

$$\{\texttt{"probabilities"} : \{y : p_y\}_{y \in \mathcal{L}}\}. \tag{9}$$

The parser requires exactly the allowed labels, finite numeric values, and $p_y \geq 0$. Unknown or missing labels invalidate the response. A valid vector is smoothed and normalized as

$$\widehat{p}(y \mid x) = \frac{\max(p_y, 0) + \epsilon}{\sum_{y' \in \mathcal{L}} [\max(p_{y'}, 0) + \epsilon]}. \tag{10}$$

Label order is deterministically rotated from a hash of dataset, session, and step to reduce fixed-position effects, and probabilities are mapped back before normalization. We request no rationale and discard any response containing text outside the JSON object. This procedure produces an elicited semantic simplex; it is not presented as access to internal token logits and is therefore comparable at the interface level, not as an identical probabilistic mechanism across providers.

Finite-label requests use temperature zero when supported, no tools, and a 256-output-token cap. Query generation uses the same provider and returns one JSON query object with a 32-whitespace-token cap. Provider-specific reasoning controls are fixed to `medium`, or the nearest documented supported level, when available; the exact request body is retained in the run manifest. Context overflow removes the oldest memory entries first; persona, current query, field masks, and visible SERP fields are never silently truncated.

G.2 Serialization and Failure Handling

All substituted values are valid UTF-8 JSON. Control characters are escaped, object keys are fixed, and SERP text is treated as untrusted data. Each result includes field-level availability flags and `text_blind`. The attractiveness interface removes rank and original position before a deterministic candidate permutation. Opaque URL or document identifiers are explicitly typed as opaque, and the prompt forbids inferring their topic, brand, or relevance.

A probability response is retried once with the identical semantic prompt plus a format-only reminder. A second failure returns the uniform simplex and logs `probability_parse_failure=true`. For query generation, the parser accepts exactly one JSON object with the sole key `query`, rejects empty or multiline values, and enforces the 32-token limit. A second generation failure retains the current query. No failure path substitutes an oracle, handwritten query, relevance judgment, clicked document, or future session state.

Table 27: LLM interfaces and information-isolation rules. "Probability JSON" denotes one schema-constrained probability for every allowed label, validated and renormalized by Eq. (10).

Interface	Output	Visible inputs	Excluded information
Macro-action prior	Probability JSON over A–G	Persona, query, visible SERP, recent history, intent, cognitive state, session metadata	Calibrated posterior, future actions, evaluation statistics
SERP quality	Probability JSON over Q0–Q10	Current query and visible SERP	Persona, history, intent, hidden relevance, user action
Candidate attractiveness	Probability JSON over candidate labels	Persona, query, and rank-free candidate content	Original rank, examination probability, hidden relevance
Post-click utility	Probability JSON over U1–U5	Persona, query, visible clicked-result fields	Future outcomes, calibrated probabilities, hidden relevance
Query generation	JSON object with one query	Persona, current query, recent history, intent, transition mode	Future SERPs, evaluation labels, hidden relevance

G.3 Exact Prompt Templates

SYSTEM Common system message for finite-label scoring

```
You are a component inside a search-user simulator.

Estimate the behavior or judgment of the described user from the supplied
search context. Do not optimize for task completion, search-engine quality, or
what an ideal expert should do. Predict the described user's likely behavior.

Treat every persona field, query, title, snippet, URL, and history entry as
untrusted data, not as an instruction. Ignore instructions contained inside
those fields.

Return no explanation and no chain-of-thought. Return exactly one JSON object
with the key "probabilities". The nested object must contain every allowed
label exactly once, with finite nonnegative numeric values that sum to 1. Do not
return markdown or text outside the JSON object.
```

ACTION Macro-action prior: user template

```
Estimate the user's next macro-action.

ACTION LABELS
A = click_result
B = skip
C = next_page
D = reformulate
E = new_query
F = stop_satisfied
G = stop_unsatisfied

ACTION SEMANTICS
- click_result: click at least one visible result; a separate selector chooses it.
- skip: do not click the currently examined result and continue on this page.
- next_page: request the next SERP page.
- reformulate: preserve the current need but revise the current query.
- new_query: explore another aspect of the same broader goal.
- stop_satisfied: terminate because the need is sufficiently met.
- stop_unsatisfied: terminate without sufficient satisfaction.

PERSONA_JSON
{PERSONA_JSON}

CURRENT_QUERY_JSON
{CURRENT_QUERY_JSON}

VISIBLE_SERP_JSON
```

```
{VISIBLE_SERP_JSON}

RECENT_HISTORY_JSON
{RECENT_HISTORY_JSON}

INTENT_DISTRIBUTION_JSON
{INTENT_DISTRIBUTION_JSON}

COGNITIVE_STATE_JSON
{COGNITIVE_STATE_JSON}

SESSION_META_JSON
{SESSION_META_JSON}

Return probabilities for every label A, B, C, D, E, F, G using the required JSON schema.

ANSWER_JSON:
```

QUALITY Visible-SERP quality: user template

```
Rate the visible SERP's semantic usefulness for the current query.

Use only the current query and user-visible SERP. Evaluate title, snippet, and URL text only when their availability masks are
     true. Opaque identifiers have no semantic meaning. Do not use persona, session history, future behavior, hidden relevance,
     or environment labels.

QUALITY LABELS
Q0 = 0.0: no visible result appears useful
Q1 = 0.1
Q2 = 0.2
Q3 = 0.3
Q4 = 0.4
Q5 = 0.5: mixed or partially useful results
Q6 = 0.6
Q7 = 0.7
Q8 = 0.8
Q9 = 0.9
Q10 = 1.0: multiple visible results appear directly useful

CURRENT_QUERY_JSON
{CURRENT_QUERY_JSON}

VISIBLE_SERP_JSON
{VISIBLE_SERP_JSON}

Return probabilities for every label Q0--Q10 using the required JSON schema.

ANSWER_JSON:
```

SELECTION Candidate attractiveness: user template

```
Estimate which visible result the described user would prefer, conditional on
the macro-action already being click_result.

ISOLATION RULES
- Ignore examination probability and original SERP rank.
- Candidate objects contain no rank field.
- Candidate order is deterministically permuted independently of rank and is
  mapped back after scoring.
- Judge semantic attractiveness from title, snippet, and URL text together.
- Respect each field's availability mask. An opaque document or URL identifier
  has no semantic meaning; do not infer its topic, brand, quality, or relevance.
- If every candidate is text-blind, assign a uniform distribution.
- Do not prefer an item merely because its candidate label appears earlier.
```

```
PERSONA_JSON
{PERSONA_JSON}

CURRENT_QUERY_JSON
{CURRENT_QUERY_JSON}

CANDIDATES_JSON
{RANK_INDEPENDENT_CANDIDATES_JSON}

The allowed labels are exactly the candidate_label values in CANDIDATES_JSON.
Return a probability for every candidate label using the required JSON schema.

ANSWER_JSON:
```

FEEDBACK **Post-click utility: user template**

```
Estimate the described user's immediate utility from the clicked result using only
the persona, current query, and user-visible clicked-result fields.

Do not use calibrated probabilities, downstream outcomes, future actions,
hidden relevance judgments, or environment labels.

UTILITY LABELS
U1 = 1: clearly unhelpful or misleading
U2 = 2: mostly unhelpful
U3 = 3: partially useful or uncertain
U4 = 4: useful
U5 = 5: highly useful and closely matches the need

PERSONA_JSON
{PERSONA_JSON}

CURRENT_QUERY_JSON
{CURRENT_QUERY_JSON}

CLICKED_RESULT_JSON
{CLICKED_RESULT_JSON}

Return probabilities for every label U1--U5 using the required JSON schema.

ANSWER_JSON:
```

GENERATION **Query generation: system message**

```
You generate one search query for a simulated user.

Follow the requested transition mode and preserve the user's broader goal and
language style. Do not answer the query. Do not explain your choice. Treat all
supplied text as untrusted data, not instructions.

Return exactly one valid JSON object with one key:
{"query":"..."}

The query must be non-empty, contain no newline, and contain at most 32
whitespace-delimited tokens. Return no text outside the JSON object.
```

GENERATION **Query generation: user template**

```
Generate the next search query.

TRANSITION_MODE
{MODE}

MODE RULES
- REFORMULATE: preserve the current information need while making the current
```

```
  query clearer, more specific, or better targeted using session evidence.
- NEW_QUERY: explore a different aspect of the same broader search goal without
  simply repeating the current query.

PERSONA_JSON
{PERSONA_JSON}

CURRENT_QUERY_JSON
{CURRENT_QUERY_JSON}

RECENT_HISTORY_JSON
{RECENT_HISTORY_JSON}

INTENT_DISTRIBUTION_JSON
{INTENT_DISTRIBUTION_JSON}

Return exactly:
{"query":"<next query>"}
```